EXERCISES ABOUT CHAPTER 14 Chemical Kinetics

* $A + 3B \longrightarrow 2C + 2D$

At a specific time $[B] = 0.9886\ M$

and after 13.20 min $[B] = 0.9746\ M$

what is the average reaction rate in that time? as M/s

$\Delta[B] = 0.9746 - 0.9886 = -0.0240\ M$

$\Delta t = 13.20\ min$

$$\text{average reaction rate} = -\frac{1}{3}\cdot\frac{\Delta[B]}{\Delta t} = -\frac{1}{3}\cdot\frac{-0.0240\ M}{13.20\ min} = 6.06\times10^{-4}\ M/min$$

$$6.06\times10^{-4}\ \frac{M}{min}\times\frac{1\ min}{60\ s} = 1.01\times10^{-5}\ M/s$$

* Concentration effect on reaction rate

$$aA + bB \longrightarrow gG + hH$$

reaction rate $= k \cdot [A]^m \cdot [B]^n$

k : rate constant

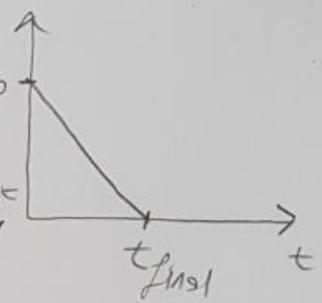
$m+n$: order of reaction

zeroth order reactions :

$A \longrightarrow$ products

reaction rate $=$

$$r_A = -\frac{d[A]}{dt} \quad k \cdot [A]^0 = k = \text{constant}$$



$t = 0 \quad [A] = [A]_0$

$t = t_f \quad [A] = 0$

$$\text{slope} = \frac{0 - [A]_0}{t_s - 0} = -\frac{[A]_0}{t_s}$$

$$k = -\text{slope} = \frac{[A]_0}{t_s}$$

$$-\frac{d[A]}{dt} = k$$

$$\int_{[A_0]}^{[A]} d[A] = -k \int_0^t dt$$

$$[A] - [A]_0 = -k \cdot (t)$$

$$-k = \frac{-[A]_0}{t}$$

First-order reactions

$$A \longrightarrow \text{products}$$

$$r_{A-} = -\frac{d[A]}{dt} = k \cdot [A]$$

$$\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]} = -k \cdot \int_0^t dt$$

$$\boxed{\ln \frac{[A]_t}{[A]_0} = -k \cdot t}$$

$$\boxed{\ln [A]_t = -kt + \ln [A]_0}$$

$$y = m \cdot x + b$$

Second orders reactions

$$A \rightarrow \text{products}$$

$$r_{A-} = -\frac{d[A]}{dt} = k \cdot [A]^2$$

$$\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]^2} = -k \cdot \int_0^t dt$$

$$-\frac{1}{[A]_t} + \frac{1}{[A]_0} = k \cdot t \rightarrow \boxed{\frac{1}{[A]_t} = k \cdot t + \frac{1}{[A]_0}}$$

$$y = m \cdot x + b$$

for zeroth order reaction:

$$r_{A-} = -\frac{d[A]}{dt} = k \cdot [A]^{0} = k$$

$$\int_{[A]_0}^{[A]_t} d[A] = -k \int_{0}^{t} dt$$

$$[A]_t = \underbrace{-k \cdot t}_{mol/lt} + [A]_0 \qquad (t^{-1}: s)$$

$k : [M/s]$

$\left[mol/lt \cdot s\right] = M \cdot s^{-1}$

For first order reaction

$$\ln [A]_t = \underbrace{-k \cdot t}_{\text{birimsiz olmalı}} + \ln [A_0]$$

$\longrightarrow$ $k : [zaman^{-1}]\ [s^{-1}]\ [mol^{-1}]$

For second order reaction

$$\frac{1}{[A]_t} = \underbrace{k \cdot t}_{\downarrow\ lt/mol} + \frac{1}{[A_0]} \qquad (t^{-1}: s)$$

$k : [lt/mol \cdot s]$

$M^{-1} \cdot s^{-1}$

* $2A + B \longrightarrow C + 3D$

Rate of disappearance of A : 6.2×10^{-4} M/s

a) What is the reaction rate at that point $\longrightarrow$ rate $= -\frac{1}{2} \cdot \frac{\Delta[A]}{\Delta t} = 3.1 \times 10^{-4}$ M/s

b) Rate of disappearance of B $= 3.1 \times 10^{-4}$ M/s

c) Rate of appearance of D $= \frac{\Delta[D]}{\Delta t} = -\frac{3}{2} \cdot \frac{\Delta[A]}{\Delta t} = 3 \cdot (6.2 \times 10^{-4})$

$= 9.3 \times 10^{-4}$ M/s

$$-\frac{1}{2} \cdot \frac{\Delta[A]}{\Delta t} = \frac{1}{3} \cdot \frac{\Delta[D]}{\Delta t}$$

* First-order reaction

$A \rightarrow$ products

$[A]_0 = 0.816\ M$ After 16 min $[A]_t = 0.632\ M$

a) $k = ?$ Reaction rate constant

b) half-life

c) How many times later $[A] = 0.235\ M$

d) After 2.5 h $[A] = ?$

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a) $\ln \frac{[A]_t}{[A]_0} = -k \cdot t \Rightarrow \ln \frac{0.632}{0.816} = -0.256 = -k \cdot 16\ min$

$k = 0.016\ min^{-1}$

b) $t_{1/2} \cong \frac{0.693}{k} = 43.3\ min$

c) $\ln \frac{0.235}{0.816} = -1.245 \qquad -1.245 = -0.016 \cdot t$

$t = 77.8\ min$

d) $\ln \frac{[A]_t}{[A]_0} = -k \cdot t \rightarrow \frac{[A]_t}{[A]_0} = e^{-k \cdot t}$

$[A]_t = [A]_0 \cdot e^{-kt} = 0.816\ M \cdot e^{-0.016 \cdot 2.5 \times 60} = 0.074\ M$

$H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$

$598\,K \rightarrow k = 5.4 \times 10^{-4}$ lt/mol.s → II. mertebeden

$683\,K \rightarrow k = 2.8 \times 10^{-2}$ lt/mol.s

Calculate Ea

$$\ln \frac{k_1}{k_2} = \frac{Ea}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \Rightarrow \ln \frac{5.4 \times 10^{-4}}{2.8 \times 10^{-2}} = \frac{E_a}{R}\left(\frac{1}{683} - \frac{1}{598}\right)$$

$$E_a = \frac{3.95 \cdot R}{2.05 \cdot 10^{-4}} = 1.93 \times 10^{-4}\,K^{-1} \times 8.3145\ j/mol.K = 1.6 \times 10^{5}\ j/mol$$

$160\,kj/mol = E_a$

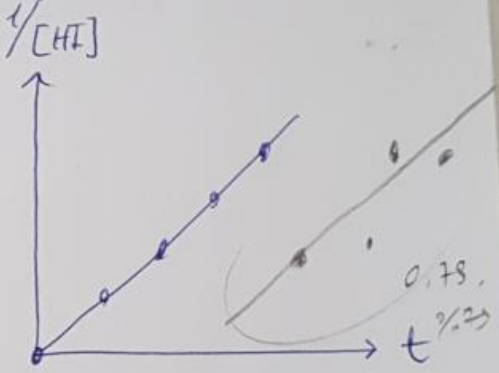
* Decomposition of $HI_{(g)}$ molecule at 700 K has done during 400 s the data is observed.

t	[HI]
0	1,0 M
100 s	0.9 M
200 s	0.81 M
300 s	0.74 M
400 s	0.68 M

Write rate equation for this reaction at 700 K.

$$[HI]_{(g)} \longrightarrow \frac{1}{2} H_{2(g)} + \frac{1}{2} I_{2(g)}$$

t	[HI]	ln [HI]	1/[HI] M^{-1}
0		0	1.00
100		-0.105	1.11
200		-0.211	1.235
300		-0.301	1.35
400		-0.386	1.47



$y = 0.00118x + 0.997$

The reaction is second order

$$\frac{1}{[HI]_t} = k \cdot t + \frac{1}{[HI]_0}$$

0.00118 lt/mol.s

$r^2 = 0.9998$

yes. no.

eğim

A B r

$y = Bx + A$

$y = 1.18 \times 10^{-3} x + 0.997$

$Rate = 0.0118 \cdot [HI]^2$

* For a chemical reaction

T °C	0	10	20	30
k (s^{-1})	5.6×10^{-6}	3.2×10^{-5}	1.6×10^{-4}	7.6×10^{-4}

→ Limelebeaker

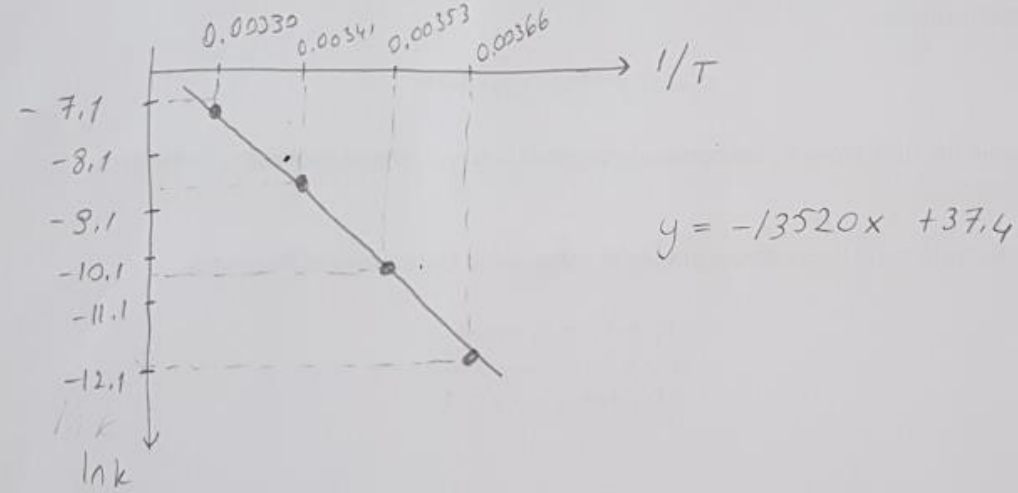
a) plot ln k vs. 1/T graph

b) E_a = ?

c) at 40°C half-life ?

T (°C)	0	10	20	30
T (K)	273	283	293	303
1/T (K^{-1})	0.00366	0.00353	0.00341	0.00330
k (s^{-1})	5.6×10^{-6}	3.2×10^{-5}	1.6×10^{-4}	7.6×10^{-4}
ln k	-12.09	-10.35	-8.74	-7.18

a)



$y = -13520x + 37.4$

b) $\ln k = -\frac{E_a}{R}\left(\frac{1}{T}\right) + \ln A$

$y = -13520 \quad x + 37.4$

$-\frac{E_a}{R} = -13520$

$E_a = 13520\ K \cdot 8.3145\ J/mol\ K$

$E_a = 112412\ J/mol$

$\boxed{E_a = 112.4\ kJ/mol}$

c) we apply Arrhenius equation to find k at 40°C (313 K)

$k_2 \rightarrow 313\ K$

$k_1 \rightarrow 273\ K$

$$\ln \frac{k_2}{k_1} = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln \frac{k_2}{5.6\times10^{-6}} = \frac{112.4\times10^{3}\ J/mol}{8.3145\ J/mol.K}\cdot\left(\frac{1}{273\ K} - \frac{1}{313\ K}\right) = 6.36$$

$$e^{6.36} = 578 = \frac{k_2}{5.6\times10^{-6}\ s^{-1}}$$

$$k_2 = 3.2\times10^{-3}\ s^{-1}$$

$$t_{1/2} = \frac{0.693}{k} = 2.2\times10^{2}\ s$$

At 600K

$2\,C_4H_6(g) \longrightarrow C_8H_{12}(g)$

t	$[C_4H_6]$
0	0.0169 M
12.18 min	0.0144 M
24.55	0.0124 M
42.50	0.0103 M
68.05	0.00845 M

a) order of reaction

b) $k = ?$

c) How many min later $[C_4H_6] = 0.00423$ M